

SURESH.M

Head - Technical & NPD

Embedded Systems | Power Electronics

EV Battery (LV&HV) | BMS | EV Charger | HUPS | Inverter | SMPS | DC-DC Converter

+91-9885770588 | msuresh29@gmail.com



PROFESSIONAL SUMMARY

New Product Development (NPD) leader in Embedded Systems & Power Electronics domains with 21+ years of experience in Battery Management Systems (BMS), high-voltage battery packs, high density AC-DC rectifiers/Chargers/ SMPS, Home UPS, Higher KVA Inverters, DC-DC converters, Battery chargers & Charge Discharge circuits and intelligent Energy Storage Solutions (ESS). Strong experience in Li-ion battery safety systems, embedded firmware, CAN communication, charger integration, remote diagnostics and battery pack electronics. Experienced in developing scalable BMS platforms for multiple EV OEMs including swappable and fixed battery architectures.

CORE COMPETENCIES & TECHNICAL SKILLS

• Battery Management Systems (BMS) • High density AC-DC Rectifiers/Chargers/SMPS • Home UPS, Higher KVA Inverters • DC-DC converters • Battery chargers & Charge Discharge circuits • High Voltage Battery Systems • Li-ion Batteries (LFP / NMC) • Battery Safety Systems • Thermal Protection Systems • Cell Balancing & Diagnostics • SOC / SOH Estimation • Swappable Battery Architecture • Power Electronics & Energy Storage • HV-LV System Integration • Battery Pack Integration • Battery Telematics & Remote Diagnostics	• CAN / RS485 / MODBUS / SPI / I2C • MODBUS / Bluetooth / GPS& GPRS / RF transceivers • RTC / ADC / external Flash integration • Charger Communication Protocols • Standards: AIS-156, AIS-004, AIS038, ISO26262 • Product Development & Commercialization • Technical Leadership & Cost Optimization • Embedded Systems & Firmware • Embedded C Programming • Schematic PCB Design in Altium • Microcontrollers & DSC: 8-bit, 16-bit, 32-bit • Microcontrollers: NXP, ST, Freescale, TI • IDE: Kinetis(KDS), Code Warrior, Code Composer • Studio(CCS), AVR studio, Keil • EV Powertrain Support Systems • Functional Safety Awareness
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KEY ACHIEVEMENTS

- Received 'Company of the Year – Battery Management System' Award in 2023 from E-Mobility+.
- Led development and deployment of BMS platforms for multiple EV OEMs.
- Developed smart BMS solutions for swappable and fixed EV battery platforms.
- Delivered telecom lithium battery solutions for Reliance Jio, Airtel and Indus Towers.
- Developed high-efficiency power electronic systems for Railway and Telecom applications.
- Successfully handled products from concept development to field deployment.

PROFESSIONAL EXPERIENCE

TriVolt New Energy Pvt. Ltd. | Head – Technical & NPD | Aug 2024 – Present

- Leading New Product Development activities for Lithium battery platform and energy storage products.
- Driving product architecture, BMS development, EV Charger sub components.
- Managing Embedded systems, Power Electronics and customer technical support.
- Supporting next-generation Energy Storage Solution platform development and commercialization.

Amara Raja Power Systems Ltd. | Sr. Manager – Technology & Product Dev. | Dec 2005 – Aug 2024

Led multiple strategic product development programs in EV domain, Telecom energy storage and Railway power products such as BMS, EV Charger, Home UPS, Higher KVA Inverters, DC-DC Converters and Charge Discharge(C/D) formation circuits.

EV Battery Management System (BMS) Development

Clients: Piaggio, Tork Motors, Omega Seiki, Baxy, Reevanta, MLR Auto

- Led architecture and development of smart BMS platforms for EV battery packs ranging from 1.5KWh to 10.2KWh for NMC & LFP chemistry batteries
- Designed scalable BMS solutions compatible with swappable and non-swappable EV platforms.
- Implemented battery safety protections including over-voltage, over-current and thermal protection mechanisms.
- Integrated CAN, RS485, Bluetooth and GPS/GPRS communication systems for diagnostics & connectivity.
- Developed battery monitoring, fault logging and remote diagnostic capabilities.
- Led embedded firmware, hardware design and PCB development activities.
- Supported OEM integration and field deployment activities.

Telecom Lithium Battery BMS Development

Clients: Reliance Jio, Airtel, Indus Towers

- Developed intelligent BMS solutions for Telecom lithium battery backup systems.
- Implemented CAN and RS485 interfaces for charger and rack communication.
- Developed data logging and remote monitoring functionalities.
- Improved battery reliability and operational safety for telecom deployments.

Integrated Power Supply Systems – Indian Railways

- Led development of next-generation Integrated Power Supply (IPS) systems for Railway signaling applications.
- Developed communication-enabled rectifiers, inverters and DC-DC converter systems.
- Integrated intelligent diagnostics and remote monitoring systems.
- Supported compliance with Indian Railway specifications and reliability standards.

48V / 100A SMPS (5.6kW) – Telecom Application

- Developed high-efficiency Telecom SMPS systems with Active Front End Rectifier architecture.
- Worked on digital control algorithms and embedded firmware for high-power converter systems.
- Achieved system efficiency up to 92% with low harmonic distortion.

Sine Wave Home UPS

- Developed using sine code PWM algorithm in 16-bit.
- Freescale Digital Signal Controller(DSC).
- Ratings: 400VA, 650VA, 800VA and 1400VA
- H-Bridge topology is used to convert the low voltage 12V DC to 230V AC pure sine
- Used Freescale Digital Signal Controller (DSC) 56F8014.

Higher KVA Inverter

Client : Indian Railways

- Developed inverter for Indian Railways by fulfilling RDSO specification for signaling purpose.
- Product ratings: 500VA, 1.5KVA, 2.0KVA and 3KVA
- H-Bridge topology is used to convert the low voltage 110V DC to 230V AC pure sine
- Used Freescale Digital Signal Controller (DSC) 56F8014

Binary Rely Intelligent Converter(BRIC)

Client : Indian Railways

- Convert from 110VDC to specified DC output voltage such as 12-40V/10A, 100-120V/3A, 3-12V/5A etc.,
- Used H-Bridge topology and Phase Shift based PWM controller plays major role in DC-DC converter to achieve Zero Voltage Switching(ZVS).
- The basic Voltage reference, Current reference generation and communication to Control Supervisory Unit(CSU) by Digital Signal Controller MC56F8014.

EARLIER EXPERIENCE

- Embedded Engineer – R&D | Sifco Processor (Power Synthesis Pvt. Ltd.) | May 2005 – Nov 2005
- Embedded Engineer – R&D | S.R. Electronics | Dec 2004 – Apr 2005

EDUCATION

M.Sc. (Electronics) – Sri Venkateswara University, Tirupati | 2004 | 72%

AWARDS & RECOGNITION

- Company of the Year – Battery Management System Award | 2023

OTHER INFORMATION

- Nationality : Indian
- Present Address : Tirupati, Andhra Pradesh – 517501
- e-mail ID : msuresh29@gmail.com
- Mobile No : +91-9885770588